I. Hyperparameters and Their Role

Q1: What is the definition of a hyperparameter in Machine Learning models?
A1: A **hyperparameter** is a parameter **set before training** that controls the learning process but is not learned from the data.

Q2: Provide three examples of common hyperparameters in a Neural Network (NN).
A2: Learning rate, number of hidden layers, number of neurons per layer.

Q3: Name one key hyperparameter for Ridge Regression or LASSO.
A3: The **regularization strength** ((\lambda) or (\alpha)).

Q4: In SVM, what three types of parameters are typically hyperparameters?
A4: Kernel type, regularization parameter (C), and kernel-specific parameters (e.g., gamma for RBF).

Q5: What two configuration variables in Random Forest are hyperparameters?
A5: Number of trees (n_estimators) and **maximum depth of trees (max_depth)**.

Q6: Why must hyperparameters be specified before model training?
A6: Because they **control model complexity and learning behavior**, which affects training outcomes.

II. Grid Search Cross-Validation (Grid Search CV)

Q7: What is Grid Search CV for hyperparameter tuning?
A7: It is an **exhaustive search over a predefined grid of hyperparameter values**, using cross-validation to evaluate performance.

Q8: How is the optimal set of hyperparameters determined?
A8: By selecting the combination that **achieves the best cross-validated performance metric**.

Q9: What is the main drawback of Grid Search CV?
A9: It is **computationally expensive**, especially for large hyperparameter spaces.

Q10: For what size of hyperparameter search space does Grid Search work best?
A10: Small to moderate-sized search spaces.

Q11: If the best hyperparameters are determined, what is the next step before final testing?
A11: Retrain the model on the training data using the optimal hyperparameters.

Q12: Once the final model is trained, what is the test set used for?
A12: To **evaluate the model's performance on unseen data**.

III. Random Search Cross-Validation (Random Search CV)

Q13: What is the fundamental difference between Random Search CV and Grid Search CV?
A13: Random Search **samples hyperparameters randomly**, while Grid Search examines **all combinations exhaustively**.

Q14: What makes Random Search CV more efficient?
A14: It **covers the search space faster** and is better for **high-dimensional or large hyperparameter spaces**.

Q15: What is the main disadvantage of Random Search CV?
A15: It **might miss the true optimal combination** due to random sampling.

IV. Optimization Without Cross-Validation

Q16: When might a fixed validation set be used instead of CV?
A16: When **speed is critical** and a single validation set suffices for hyperparameter evaluation.

Q17: What is the primary motivation for skipping CV?
A17: Faster evaluation, avoiding repeated model training across folds.

Q18: When is using a fixed validation set not recommended?
A18: When the dataset is **small or variable**, as it can produce **biased performance estimates**.

V. Bayesian Optimization

Q19: What is the goal of Bayesian Optimization in hyperparameter tuning?
A19: To **find the optimal hyperparameters efficiently** using a probabilistic model.

Q20: What type of model does Bayesian Optimization create?
A20: A **surrogate probabilistic model** of the ML model's cross-validation performance.

Q21: How is the probabilistic model used to select the next hyperparameter choice?
A21: By using an **acquisition function** to balance **exploration and exploitation**.

Q22: How does Bayesian Optimization compare to Grid/Random Search in efficiency?
A22: It is **more sample-efficient**, requiring **fewer evaluations** to find good hyperparameters.

Q23: Name two libraries that implement Bayesian Optimization.
A23: Hyperopt and **BayesianOptimization** (Python libraries).

VI. General Concepts and Other Methods

Q24: Why is hyperparameter tuning typically done with Cross-Validation?
A24: To **ensure robust performance estimates** and prevent overfitting to a single validation set.

Q25: Which CV type is preferred in hyperparameter tuning?
A25: k-Fold Cross-Validation, due to its balance of efficiency and robustness.

Q26: Name two other approaches for hyperparameter optimization.
A26: Genetic Algorithms and **Simulated Annealing**.

Q27: What does a "small search space" refer to?
A27: A **limited set of hyperparameter values** to evaluate.

Q28: Why re-train the model on the entire training dataset after finding the best hyperparameters?
A28: To **maximize data usage** and obtain the **best-performing final model**.

Q29: What is the term for a search technique that eliminates less-performing hyperparameter configurations?
A29: Successive Halving or **Hyperband**.

Q30: How is the best hyperparameter performance typically quantified?
A30: Using **cross-validated metrics** such as **accuracy, RMSE, or F1-score**.